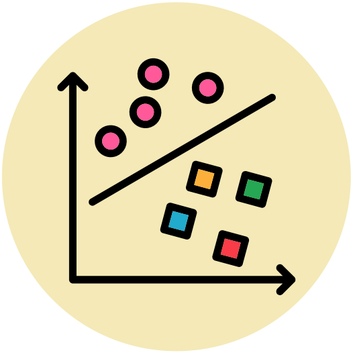

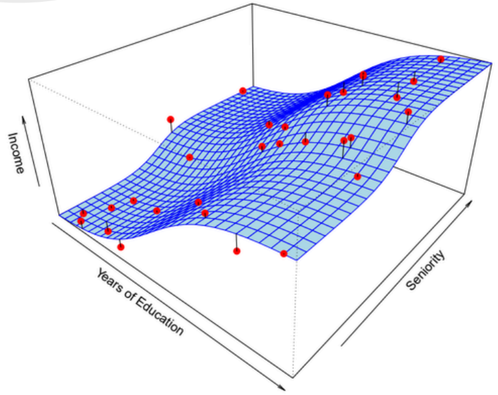

KAZAKH-BRITISH TECHNICAL UNIVERSITY

School of IT and Engineering

MACHINE LEARNING

LECTURE #5

Number of credits: 3 (2/0/1)
Course code – CSCI3234
MS Teams team code – **y4ntwlz**

Adilet Yerkin, MS in Engineering, Senior Lecturer
a.yerkin@kbtu.kz

CLASS IMBALANCE

In real life, machine learning rarely deals with perfectly balanced datasets.
Most textbooks teach ML using datasets like:

- 50% cats vs 50% dogs
- 50% spam vs 50% not spam

But in real applications — finance, medicine, cybersecurity — the interesting events are rare.
Examples:

- Fraud transactions → < 1%
- Cancer detection → < 0.5%
- Equipment failure → < 0.1%
- Customers leaving a bank → 5–15%

This situation is called class imbalance.

CLASS IMBALANCE

Class imbalance occurs when one class appears much more frequently than another in the training dataset.
Example: Cancer Detection
Imagine we have 1,000,000 X-ray images:

Class	Count
Normal	1 000 000
Cancer	100

The model sees mostly normal lungs.
So it learns: “If I always predict NORMAL — I am correct 99.99% of the time.”

This model looks accurate but is useless.

IMBALANCE ALSO EXISTS IN REGRESSION

Not only classification suffers.
Consider hospital bill prediction:

- Most bills = $200 – $500
- Rare bills = $20,000+

Predicting $300 instead of $600 → acceptable
Predicting $10,000 instead of $20,000 → catastrophic

So the goal is not average accuracy — the goal is performance on rare expensive cases.

WHY IMBALANCE BREAKS ML ?

Machine learning algorithms assume balanced information.
When imbalance exists, learning becomes difficult for three main reasons.

Problem 1 — Insufficient Signal
The model rarely sees minority examples.
So it cannot learn their pattern.
This becomes a few-shot learning problem.
Worst case: If the rare class never appears in training → the model assumes it doesn't exist.

A machine learning model learns by observing repeated patterns.
If a pattern appears frequently → the model forms a stable representation.
If a pattern appears rarely → the model cannot generalize it.

WHY IMBALANCE BREAKS ML ?

Imagine teaching a child to recognize animals.
You show:

- 10,000 photos of cats
- 5 photos of giraffes

Then you test the child.
When they see a giraffe, they say:
“This is a strange looking cat.”

Not because the child is stupid —
because the child never had enough experience to understand giraffes are different.
This is exactly what happens in ML.

What Happens Mathematically

Models learn using gradient updates.
Each training example slightly changes model parameters.

If 99.9% of data is class A:
The gradients coming from class A dominate the training.

Gradients from the rare class are drowned in noise.
So the learned decision boundary ignores the minority class.

WHY IMBALANCE BREAKS ML ?

Problem 2 — Trivial Solution Trap (The Model Learns a Shortcut)

The model finds a shortcut:
Always predict majority class.
Gradient descent prefers this because accuracy is high.

Prediction Strategy	Accuracy
Always Normal	99.99%
Actually Detect Cancer	99.2%

Optimization algorithms try to minimize loss efficiently.
When imbalance exists, there is a shortcut solution:
Predict the majority class for everything
This gives very high accuracy.
So the model converges to this local optimum.

WHY IMBALANCE BREAKS ML ?

Why Gradient Descent Gets Stuck

Gradient descent improves performance gradually.
Suppose model starts predicting NORMAL for everything. Now it tries detecting one cancer case.

Result:
Accuracy slightly decreases (because more false positives).
So optimization moves backward.
Thus:
The model learns that trying is worse than not trying.
This is called a local minimum caused by imbalance.

Human Analogy

Imagine a teacher grading exams:

- 1000 easy questions
- 1 extremely hard question

A student who ignores the hard question scores 99.9%.

So students learn:
Don't attempt the difficult question.
The training objective rewards laziness.
That is what imbalance does to ML.

WHY IMBALANCE BREAKS ML ?

Problem 3 — Unequal Cost of Errors

Not all mistakes are equally bad.

Error	Consequence
Miss cancer	Life threatening
False alarm	Extra test

But default ML treats both equally.
Therefore model optimizes wrong objective.

Standard machine learning assumes:
All mistakes cost the same.
But real life does not.

WHY IMBALANCE BREAKS ML ?

Problem 3 — Unequal Cost of Errors

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CLASS IMBALANCE IS THE REAL WORLD

Error	Meaning	Consequence
False Positive	Healthy predicted sick	Extra test
False Negative	Sick predicted healthy	**Patient dies**

The loss function optimizes average error.
But business cares about weighted error.
So the model becomes statistically correct but practically useless.

HANDLING CLASS IMBALANCE

There is no single solution.

We solve imbalance at three levels:

1. Evaluation metrics
2. Data-level methods
3. Algorithm-level methods

CHOOSING CORRECT METRICS

The biggest mistake: using accuracy

Accuracy Failure Example
Dataset: 90% Normal, 10% Cancer
Model A: predicts always NORMAL
Model B: detects most cancers

Both accuracy = 90%
But Model B saves lives.

Therefore accuracy is misleading.

	Pred Positive	Pred Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

MODEL EVALUATION

Confusion Matrix

Actual / Predicted	Fraud (1)	Legitimate (0)
Fraud (1) (Positive)	**6 (TP)**	**4 (FN)**
Legitimate (0) (Negative)	**20 (FP)**	**970 (TN)**

TP = 6, FN = 4, FP = 20, TN = 970

Accuracy:

$$\frac{6+970}{1000} = 97.6\%$$

Precision:

$$\frac{TP}{TP+FP} = \frac{6}{6+20} = 0.23 = 23\%$$

Recall:

$$\frac{TP}{TP+FN} = \frac{6}{6+4} = 0.6 = 60\%$$

F1-score:

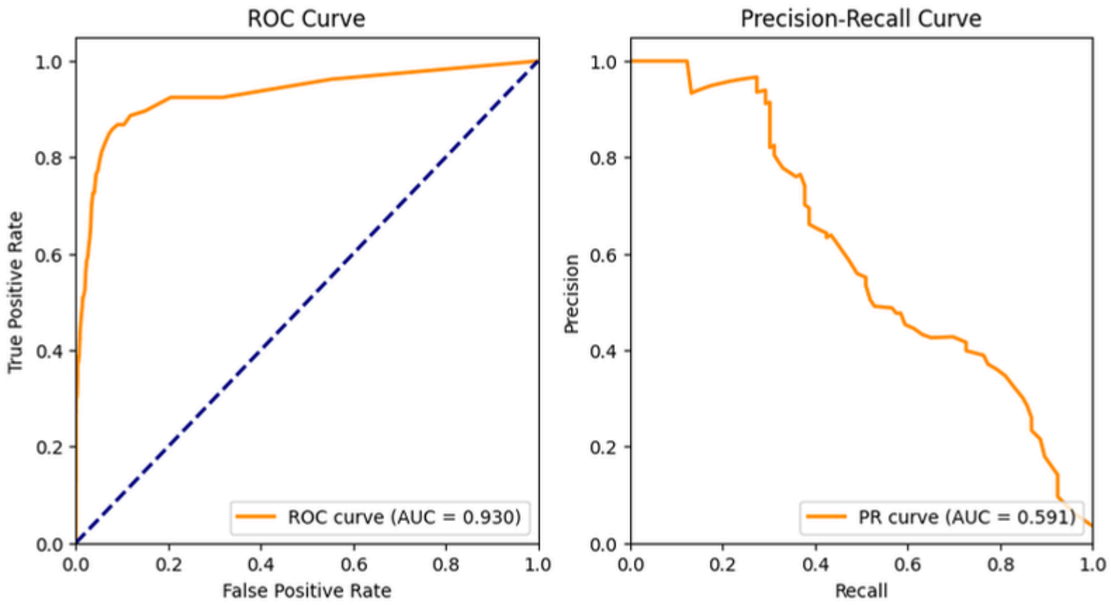
$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = 2 \times \frac{0.23 \times 0.6}{0.23 + 0.6} = 0.33 = 33\%$$

Metric	Value	Comment
Accuracy	97.60%	High, but misleading
Precision	23%	Many false alarms
Recall	60%	Detects most frauds
F1-score	33%	Balanced indicator

PRECISION-RECALL CURVE

For rare events, PR curve is more informative than ROC

ROC can look good even when model never detects positives.
PR shows performance on minority class directly.



PR-AUC answers the real question:

How well does the model find rare important events without raising too many false alarms?

ROC-AUC measures ranking ability globally.
PR-AUC measures usefulness in practice.
ROC asks: Can the model separate classes?
PR asks: When the model raises an alarm — should we trust it?
For rare events, trust matters more than separation.

PR-AUC	Meaning
0.01	Random guessing
0.10	Good model
0.30	Strong model
0.60+	Excellent model

DATA LEVEL METHODS

Goal: Show the model more minority examples.

Up to this point we understood why imbalance breaks learning.
Now the question becomes:
If the model learns from the data…
what if we carefully change the data the model sees?

This is the idea of data-level methods.

Instead of changing the algorithm, we change the experience of the model during training.
The model still thinks the world is real — but we control what examples it observes more often.

UNDERSAMPLING

Undersampling — Removing Majority Examples

Class	Count
Normal	100 000
Fraud	1000

The model mostly learns normal behavior. So we intentionally reduce normal cases:

Class	After undersampling
Normal	1000
Fraud	1000

Advantages

1. Faster training
2. Less data → fewer computations
3. Clearer decision boundary
4. The model focuses on differences instead of memorizing majority patterns
5. Works well for huge datasets

Disadvantages

We throw away real information.

The majority class often contains important diversity:

- Different behaviors
- Different patterns
- Rare normal situations

Removing them can make the model fragile.

The model may start thinking: “Anything slightly unusual must be fraud” *So false alarms increase.*

OVERSAMPLING

Instead of removing data, we duplicate minority cases.

Class	Count
Normal	100000
Fraud	1000

After oversampling:

Class	Count
Normal	100000
Fraud	100000

Intuition

If a teacher keeps asking the same difficult question again and again,
students eventually understand it.
Repetition forces learning.

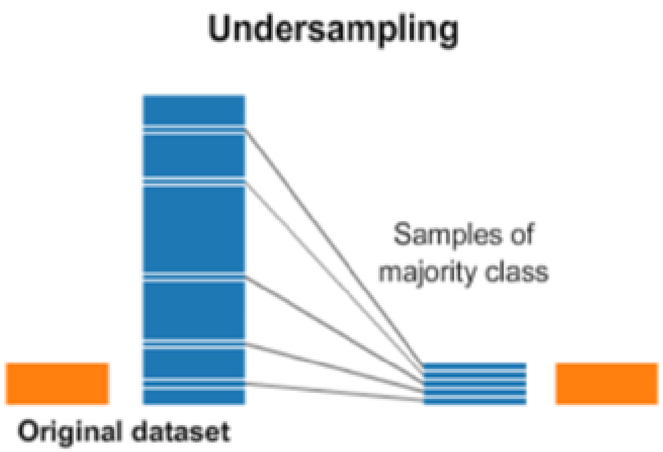
Advantages

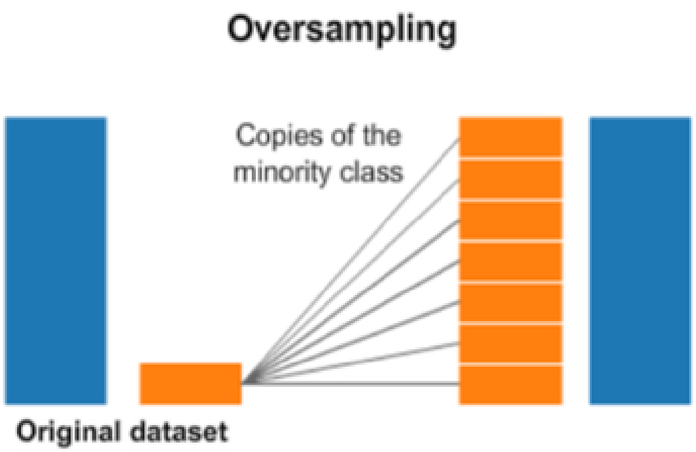
- Keeps all original data
- Very simple
- Strong baseline method

The Big Problem — Overfitting

The model memorizes instead of learning.
It doesn't learn "what fraud is".
It learns "these specific transactions are fraud".
When new fraud appears → model fails.
This is the biggest limitation of naive oversampling.

DATA LEVEL METHODS





Method	What we do
Oversampling	Add minority examples
Undersampling	Remove majority examples

SMOTE
SYNTHETIC MINORITY OVERSAMPLING

To solve memorization, we create new artificial minority samples.
Instead of copying a fraud transaction:
We create a new one between two fraud examples.

Intuition

Imagine drawing points on paper:
Fraud points cluster together.
SMOTE creates new points inside the cluster.
So the model learns the region, not individual points.

Why This Works

The model stops memorizing exact cases
and starts learning general characteristics.
This dramatically improves generalization.

Important Limitation

SMOTE works well in low or medium dimensions.
In very high-dimensional data (images, embeddings, text vectors), distance between points becomes meaningless.
Then synthetic points may become unrealistic.

ALGORITHM-LEVEL METHODS

In data-level methods, we changed the dataset so the model sees minority examples more often.
But sometimes we cannot change the data:

- Medical datasets are small
- Fraud data must stay realistic
- Regulatory environments forbid modification
- Deep learning models already require full data

So instead of changing the world the model sees, we change what the model cares about.
This leads to algorithm-level methods.

THANK YOU